Flexible Cable Simulation in NVIDIA Isaac Sim

Comparing three modelling approaches for RL-based cable manipulation

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Motivation

- **Why cables?** wire-harness assembly, cable routing, and deformable-object manipulation are common, unsolved robotics tasks.
- **The challenge:** Isaac Sim has *no native 1-D deformable (rod/cable) body* – only rigid bodies and volumetric FEM soft bodies.
- **The goal:** find the cable model that is realistic *and* fast *and* robot-coupled enough for **RL-based cable manipulation** with domain randomization.

We evaluate three approaches, each in two scenarios (free cable, and cable held by a Franka arm).

Three approaches

1. Capsule chain

rigid capsules + D6 joints; bending from Euler–Bernoulli beam theory (PhysX).

2. Cosserat rod (Warp)

1-D position-based elastic rod in NVIDIA Warp, after the *OGC* SIGGRAPH-2025 paper.

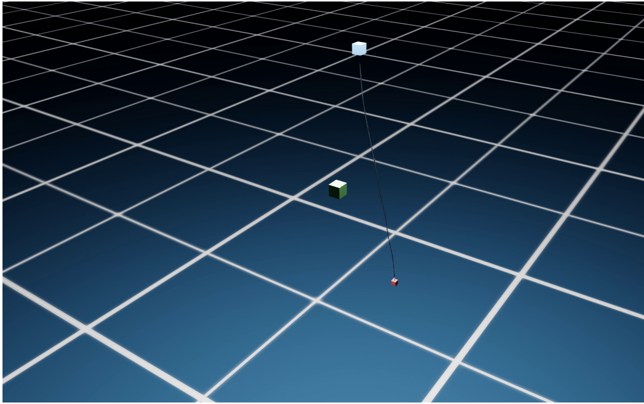
3. FEM deformable

Isaac Sim's built-in volumetric soft body (tetrahedral FEM).

Two scenarios each ⇒ **6 simulations**: (A) free hanging/swinging cable (B) cable connected to a robot manipulator.

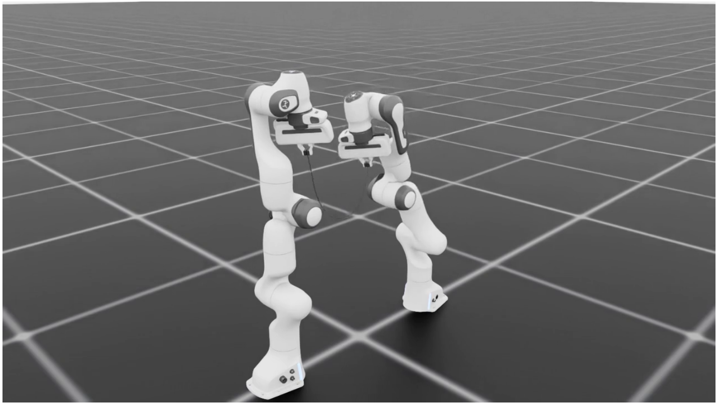
Approach 1 — Capsule chain (D6 joints)

- Rigid capsules linked by **D6 joints**; translations locked (inextensible), bending springs $k = EI/L$ from beam theory.
- PhysX **TGS** solver, up to 200 links, 240 Hz.
- Reproduces the **Govoni et al. 2025** stability sweep.
- Modes: hanging-kick and both-ends-fixed.



Video: approach1_cable_only/video.mp4 (shared folder)

Approach 1 — With robot

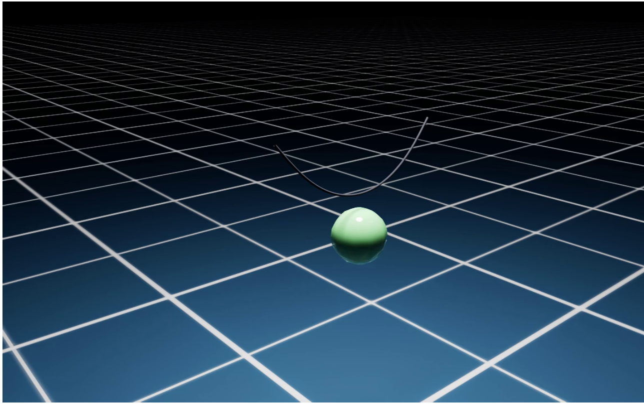


Video: approach1_robot/video.mp4 (shared folder)

Ends tied to the gripper via **ball joints** (pin) ⇒ the cable droops at the connection and **drags the second arm** (two-way coupling). Stable.

Approach 2 — Cosserat rod in NVIDIA Warp

- 1-D **Cosserat / XPBD** elastic rod: inextensible stretch + bending stiffness + gravity, solved on the GPU in **Warp**.
- Idea from the **OGC** paper (Chen et al., SIGGRAPH 2025): penetration-free contact for cloth & rods.
- Truly **thin** (no FEM fattening); **exact** node-vs-primitive collision.



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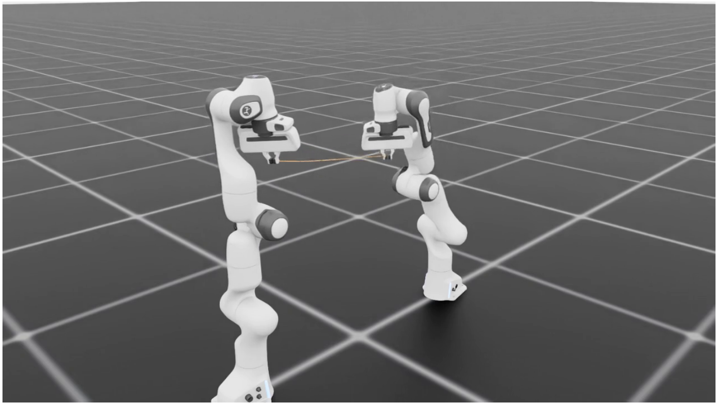
Approach 2 — With robot

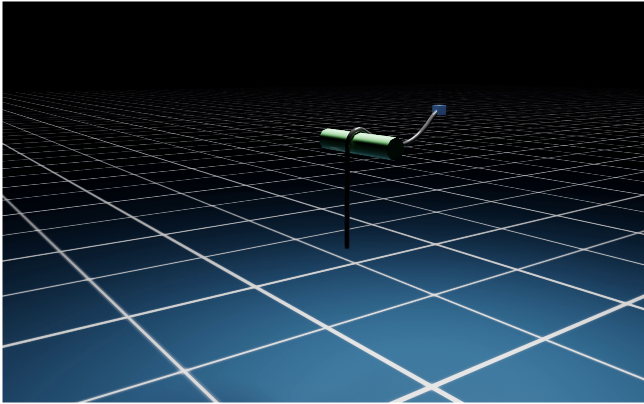


Video: approach2_robot/video.mp4 (shared folder)

End nodes **spring-coupled** to the grippers (two-way): the rod follows the arms *and* its reaction force is applied back ⇒ it **drags the follower** ($\sim$6 cm), like a real cable.

Approach 3 — FEM deformable (Isaac Sim)

- Isaac Sim's built-in **volumetric** (tetrahedral) FEM soft body.
- TPU material: $E = 40\,\text{MPa}$, $\nu = 0.48$, $\rho = 1150\,\text{kg/m}^3$.
- Rod **fattened** to 12 mm and E rescaled so it bends/weighs like the real 1.5 mm cable (voxeliser needs ≥ 3 hexes across).
- Rendered thin via a swept tube on its centreline.



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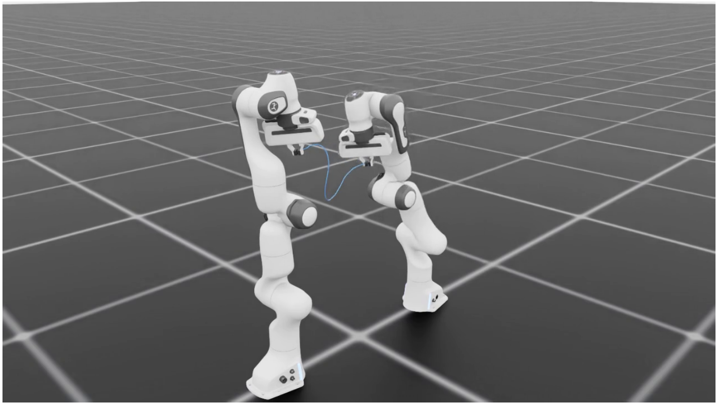
Approach 3 — With robot



Video: approach3_robot/video.mp4 (shared folder)

Attached to the gripper by a **region weld** (auto-attachment): holds and transmits force, but is **rigid/straight near the joint** and stretches.

Comparison

Criteria	**D6 capsule**	**Cosserat (Warp)**	**FEM**
Realism	Good (beam bending)	**Excellent** (1-D rod)	Fair (jelly/stretch
Stability	**Excellent**	**Excellent**	Fair
Compute cost	Low (real-time)	Low ($\sim$1.4$\times$ RT)	High ($<$0.2$\times$ RT
Ease of setup	Moderate	Moderate (custom Warp)	**Easy** (built-in)
RL readiness	**High**	**High**	Low
Robot integration	Two-way (force)	Two-way (spring-coupled)	Two-way (rough)
Domain randomization	**Easy**	Easy	Harder

RT = real time. “Two-way” = the robot feels the cable’s reaction force.

Pros and cons

1. **Capsule chain (D6 joints)**
   - + Stable over a wide parameter range; physically grounded; real-time; **two-way** robot coupling; easy domain randomization.
   - – No axial elasticity (links can't stretch); needs many links for a smooth look; free twist DOF needs damping.
2. **Cosserat rod in Warp (OGC)**
   - + Most faithful cable: thin, inextensible, no jelly; exact, penetration-free collision; GPU-fast; **two-way** robot coupling (spring + reaction force) drags the arm.
   - – Custom Warp code (not a native Isaac body); coupling to the robot is via a spring (needs tuning); no volumetric contact.
3. **FEM deformable**
   - + Most realistic volumetric deformation; handles complex geometry; built into Isaac Sim; two-way.
   - – Expensive; must be fattened (can't be thin); stretches; rough simplified collision; rigid clamp at the attachment; slow for RL.

Recommendation

For RL-based cable manipulation: the **capsule chain (D6 joints)**

- Only model that is simultaneously **stable, real-time, and two-way robot-coupled** – the robot actually feels the cable, which RL needs.
- Physically grounded (beam theory) and **easy to domain-randomize** (per-joint stiffness/damping/mass).

- Use the **Warp Cosserat rod** as a high-fidelity reference / for visualization and contact-rich studies (penetration-free, exact collision).
- Use **FEM** only when a genuine *volumetric* soft body (squashing/draping) is required – not for fast RL.

Next steps

- **RL training** of a cable-manipulation policy on the capsule model.
- **Domain randomization**: cable stiffness, damping, mass, length, friction; robot dynamics.
- **Two-way coupling** study: how the cable's reaction shapes the policy.
- **Physical validation** against a real TPU cable.
- Before arriving at Waterloo: finalize the env + reward and a first sim-to-sim transfer between the three models.

Thank you

Questions?

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